

GEOMATIKK,
NTNU

TBA4245 Satellite Navigation and GPS Plan

Coordinator, Lecturers, and Student assistant

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Lectures and exercises

Location: Lectures are held in [room L11](#). Please refer to the lecture plan for detailed information.

Schedule: Lectures take place on Tuesdays from 12:15 to 14:00 and Thursdays from 12:15 to 14:00.

Exercises and student assistant hours: These sessions are held on Fridays from 14:15 to 16:00 in [room L11](#).

This course is worth 7.5 credits (SP) and is organized as 3F + 2Ø + 7S. This corresponds to an average of three hours of lectures (F) and two hours of exercises (Ø) per week. During the first part of the semester, there will be approximately four hours of lectures per week, while in the latter part of the semester the lecture load will be reduced to about two hours per week.

Contents

GNSS code and carrier phase measurements. GNSS measurement errors and phase ambiguity resolution. GNSS satellite coordinate computations. Transformation of GNSS satellite baseline measurements into distance, azimuth and height components. Estimation of GNSS positions and heights using the least squares method, including testing and reliability analyses. An introduction to recursive least squares and Kalman filtering techniques for dynamic positioning and sensor fusion.

Canvas

Canvas will be used for announcements, assignments, lecture materials, handouts, exercises, and other course-related information. Please check Canvas regularly.

Recommended previous knowledge

Based on the course TBA4236 Theoretical Geomatics.

Learning outcome

Knowledge

After completing this course, the students should have knowledge of:

- Understanding of code- and carrier phase-based positioning, including estimation of phase ambiguities and identification of GNSS measurement error sources.
- Transformation of GNSS baselines from the Earth-Centered Earth-Fixed (ECEF) coordinate system to observations of distance and azimuth in the map projection, and height in the national height system.

- Computation and interpretation of GNSS satellite coordinates in both the ECEF and local coordinate systems
- Application of adjustment theory, including estimation, statistical testing, Kalman filtering, and reliability analysis.

Skills

After completing this course, the students should be able to:

- Estimate GNSS positions and heights from GNSS observations.
- Transform GNSS baseline observations into commonly used observations such as distance, azimuth, and height differences.
- Perform and interpret reliability tests in accordance with the Norwegian Geodata Standard issued by the Norwegian Mapping Authority.
- Understand and mitigate the effects of multipath, ionospheric, and tropospheric errors on GNSS signals, and resolve phase ambiguities in carrier-phase observations.
- Prepare for dynamic GNSS positioning and integrate GNSS data with other sensor systems.

General competence

After completing this course, the students should be able to:

- Understand the theoretical foundations for calculations on the reference ellipsoid and in map projection systems.
- Understand the principles of estimation and reliability analysis of GNSS-derived coordinates and heights.
- Demonstrate advanced knowledge of GNSS and satellite geodesy.
- Apply and communicate professional terminology accurately within the field.
- Work both independently and collaboratively, taking initiative and responsibility in professional tasks.
- Recognize connections between this discipline and other fields, and demonstrate openness to interdisciplinary approaches and cooperation.

Learning methods, activities and evaluation

Lectures. Computations and laboratory excersises. Programming. Use of existing software.

Programming, coding and implementation for the project work and obligatory exercises are included in the course.

Compulsory assignments (three assignments) must be approved in order to attend the exams. Written exam accounts for 70% of the grade and project work accounts for 30% of the grade. Each part of the assessment will be awarded a letter grade, which is then combined into a single final grade for the course.

If there is a re-sit examination of the written exam, the examination form may be changed from written to oral exam. Re-sit of any other partial assessments must be done in a semester when the course is taught.

Compulsory assignments

A combination of smaller assignments and a larger project, which accounts for 30% of the final course grade.

<i>No</i>	<i>Topic</i>	<i>Grades</i>
1	GNSS network. Baselines, from Cartesian to map projection.	Graded
2	GNSS computations.	Passed/not passed
3	Reliability analyses.	Passed/not passed
4	Geodetic main tasks	Solution given
5	Map projections and corrections.	Solution given

Example exercises

The example exercises are recommended for gaining a deeper understanding of the compulsory assignments and lecture material. They cover key topics in the course, and the accompanying solutions typically provide detailed explanations and practical applications of the concepts discussed in class. The exercises and their solutions are available on Canvas.

Course Materials and Overview

Course materials consist of textbooks and lecture notes, which are available on Canvas in PDF format. Please see the "Table of Lectures (Literature)" for details. All course materials are provided in English.

<i>No</i>	<i>Author</i>	<i>Title</i>	<i>Available at. Remarks</i>
1			
2			
3a			
4	Hofmann-Wellenhof et al	GNSS	E-book in Blackboard.
6	Gjevestad	Multiple T-test for quality control. English version.	A pdf file on Blackboard.
7			
8	Torge	Geodesy, 2012	A pdf file on Blackboard.
9	Krakiwsky	Conformal Map projections in Geodesy	A pdf file on Blackboard.

Table of lectures

The table below provides an overview of the lecture schedule. Relevant literature references are listed for each lecture to support further reading and preparation.

Date	Time	Topic	Literature	Assignments. Notes
	-			
18.8	1	Start. Administration		
20.8	2, HN	GNSS Baselines, baseline transformation, part 1	[8]: 2.5 also 6.2 and 6.3. [8]: Pages 227-229	
25.8	2, HN	GNSS Baselines, baseline transformation, part 2	[8]: 2.5 also 6.2 and 6.3 [4]: 8.2.2	Assignment 1
27.8	2, HN	GNSS Baseline transformation, part 3. Error propagation law, variance/covariance matrix	[8]: 2.5 also 6.2 and 6.3	
1.9	2, HN	GNSS Baseline transformation, part 3. Error propagation law, variance/covariance matrix	[8]: 2.5 also 6.2 and 6.3	
3.9	2, HN	Conformal map projection. Part 1.	[9]: Ch. 4, 6 [4]: Ch. 8.2.3	
8.9	2, HN	Conformal map projection. Part 1.	[9]: Ch. 4, 6 [4]: Ch. 8.2.3	
10.9	2, HN	Conformal map projection. Part 2.	[9]: Ch. 4, 6 [4]: Ch. 8.2.3	
15.9	2, HN	Conformal map projection. Part 3.	[9]: Ch. 11, 4.5, 4.6, 6.2, 6.6, 6.8	
17.9	2, HN	Conformal map projection. Part 4.	[9]: Ch. 4.2, 4.4, 6.3, 6.7, 6.8	
22.9	2, HN	GPS observations, part 1	[4]: 6	
24.9	2, HN	GPS observations, part 2	[4]: 6 and 3.4.2	Assignment 2
29.9	2, HN	GPS observations, part 2	[4]: 6 and 3.4.2	Assignment 2
1.10	2, HN	GPS Ambiguity in phase	[4]: 7.2.1, 7.2.2, 7.2.3 (214-217) and 5.2.1	
6.10	2, TAH	Reliability Analyses (RA). Overview and RA, part 1	[6]	
8.10	2, TAH	RA, part 2	[6]	Assignment 3
13.10	2, TAH	RA, part 3	[6]	
15.10	2, TAH	RA, part 4	[6]	
20.10	2, HN	GPS error. Ionosphere.	[4]: 5.3	

22.10	2, HN	GPS error. Ionosphere.	[4]: 5.3	
27.10	2, HN	GPS error. Troposphere.	[4]: 5.3	
29.10	2, HN	GPS error. Multipath.	[4]: 5.6	
3.11		???		
5.11		No lecture. Working with assignments		
10.11		No lecture. Working with assignments		
12.11		No lecture. Working with assignments		
17.11		No lecture. Working with assignments		
19.11		No lecture. Working with assignments		
??		Q&A and Exam Preparation		
22.12	15-19	The final examination accounts for 70% of the final course grade. Permitted examination aids: Category D		

Hossein Nahavandchi (HN), Trond Arve Haakonsen (TAH)